

Emotion-Based Music Recommender

Python Project Report

Natural Language Processing · Supervised Machine Learning · Desktop GUI

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Academic Year	2025 – 2026

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1. Introduction

The convergence of Artificial Intelligence, Natural Language Processing, and music technology has opened new possibilities for creating deeply personalized digital experiences. Music is one of the most universal forms of emotional expression and regulation. Research in music psychology has consistently shown that people select music based on their current emotional state — a person feeling melancholic seeks comfort in slow ballads, while someone joyful gravitates toward upbeat tracks.

Despite the proliferation of intelligent music streaming platforms such as Spotify, Apple Music, and YouTube Music, most recommendation engines primarily rely on historical listening behaviour, collaborative filtering, or manually created playlists. These approaches suffer from a fundamental limitation: they cannot adapt in real time to a user's fluctuating emotional state. A user who routinely listens to energetic workout music may be feeling anxious on a given evening and needs a completely different musical experience.

This project — the Emotion-Based Music Recommender — addresses this gap by directly detecting a user's emotional state from free-form text input using a trained machine learning classifier, and immediately mapping that emotion to a curated database of songs with playable YouTube links.

1.1 Project Context

This work was undertaken by Prabhjot Singh (SAP ID: 590023045) as part of the AI/ML curriculum at UPES (University of Petroleum and Energy Studies) for the academic year 2025– 2026, under the expert supervision of Mr. Sandeep Kumain. The project integrates theoretical machine learning concepts with practical software engineering to deliver a deployable desktop application.

1.2 Scope of the Project

- Text-based emotion detection using a supervised ML classifier.
- Nine emotion categories: happy, sad, angry, stressed, neutral, love, fear, surprise, bored.
- Dataset-driven music recommendation with YouTube links.
- Fully offline desktop application with an animated, dark-themed GUI.
- End-to-end pipeline: raw text → preprocessing → prediction → recommendation → display.

2. Abstract

This project presents a complete, end-to-end system for emotion-based music recommendation powered by Natural Language Processing and supervised machine learning. The system accepts free-form text describing a user's emotional state, classifies the emotion using a Logistic Regression model, and recommends appropriate songs from a pre-populated music dataset.

The classification pipeline is built using scikit-learn and combines TF-IDF (Term Frequency–Inverse Document Frequency) vectorisation with n-gram feature extraction (unigrams through trigrams) and a Logistic Regression classifier with balanced class weights. The model is trained on a labelled emotion dataset with an 80/20 stratified train-test split, achieving an estimated accuracy of 85–90% on the held-out test set.

Text preprocessing employs standard NLP techniques: lowercasing, URL removal, punctuation stripping, English stopword elimination via NLTK, and WordNet lemmatisation. The music recommendation component filters a CSV-based music dataset by the predicted emotion label and randomly samples up to five tracks for variety.

The user interface is a fully custom dark-themed desktop application built entirely with Python's standard library (tkinter, webbrowser, threading), requiring no external GUI dependencies. Animated emotion display, colour-coded feedback, clickable song cards, and smooth hover interactions deliver a polished user experience.

Component	Detail	Value / Technology
Classifier	Logistic Regression	scikit-learn
Features	TF-IDF (1–3 grams)	Max 15,000 features
Test Split	Stratified 80/20	random_state = 42
Accuracy	Test set performance	~85 – 90 %
Emotions	9 classes	happy, sad, angry, stressed...
GUI Framework	Tkinter (built-in)	No external dependencies
Recommendation	CSV music dataset	Up to 5 songs per emotion

3. Problem Statement

Music recommendation technology has advanced significantly over the past decade, yet a critical dimension remains largely unaddressed by mainstream platforms: the user's real-time emotional state. The following subsections outline the specific problems this project solves.

3.1 Limitation of Existing Systems

- Collaborative and content-based filtering recommend songs similar to past behaviour, not present mood.
- Platforms such as Spotify use audio features (tempo, energy) but cannot interpret the user's emotional state in natural language.
- Users must manually search for 'relaxing music' or 'angry songs' — a cognitive burden during emotionally difficult moments.

- **Pre-curated mood playlists are not dynamic and do not adapt to nuanced emotional expressions.**
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3.2 Research Gap

While emotion detection from text (sentiment analysis) is a well-studied NLP problem, its integration into a real-time, desktop-based music recommendation workflow with a polished GUI remains underexplored at the student project level. Most academic systems stop at classification accuracy and do not produce a deployable, user-friendly product.

3.3 Proposed Solution

This project solves these problems by building a lightweight, fully offline desktop application that:

1. Accepts natural language emotional input from the user (e.g., 'I feel exhausted and overwhelmed today').
2. Automatically detects the dominant emotion using a trained ML classifier.
3. Filters a music dataset by the detected emotion.
4. Presents up to five recommended songs with direct YouTube links.
5. Delivers the experience through an animated, accessible, dark-themed GUI.

3.4 Why This Problem Matters

Mental health awareness has grown significantly. Music therapy is recognised as a scientifically valid therapeutic tool. A system that can autonomously match music to a person's emotional state has genuine applications in wellness, healthcare, and everyday life. By making this technology accessible through a local desktop app, this project demonstrates how ML can be made practical, private, and impactful without cloud infrastructure.

4. Literature Review

A substantial body of research informs the design choices made in this project. The following sections review key prior works in emotion detection, music recommendation, and NLP-based classification.

4.1 Emotion Detection from Text

Early work in sentiment analysis focused on binary polarity (positive vs. negative). Ekman's foundational 1992 model proposed six universal basic emotions: happiness, sadness, anger, fear, disgust, and surprise — a framework widely adopted in NLP research. Subsequent datasets such as the SemEval Emotion Intensity Task and the GoEmotions corpus (Google, 2021) expanded emotion taxonomies to handle fine-grained multi-label classification.

For text classification tasks, TF-IDF with Logistic Regression has proven to be a strong baseline. Studies by Wang & Manning (2012) demonstrated that simple linear models with bag-of-n-gram features achieve near state-of-the-art performance on many sentiment tasks, often outperforming more complex models when training data is limited.

4.2 Music Recommendation Systems

Collaborative filtering (CF), pioneered by Amazon and Netflix, underpins most commercial recommenders. Content-based filtering analyses audio features such as tempo, key, loudness, and danceability (Spotify's audio features API). Context-aware recommendation systems began incorporating time of day, location, and activity as signals. Emotion-aware music recommendation emerged as a research domain in the 2010s, with papers demonstrating that incorporating mood labels improves user satisfaction.

A key challenge identified in the literature is the 'cold start' problem: collaborative filtering fails for new users with no listening history. Emotion-based systems sidestep this entirely by relying only on the current input text, making them effective from the very first interaction.

4.3 NLP Preprocessing Techniques

The preprocessing pipeline in this project follows established NLP best practices. Stopword removal and lemmatisation are standard steps that reduce vocabulary dimensionality without loss of semantic information. The NLTK WordNet lemmatiser normalises inflected forms (running → run, happier → happy), improving classifier generalisation. Sublinear TF scaling ($\log(1 + \text{TF})$) dampens the effect of very frequent terms, a technique shown to improve performance on short text classification tasks.

4.4 Summary and Positioning

This project synthesises lessons from the above literature: TF-IDF + Logistic Regression for efficient text classification, balanced class weights for imbalanced emotion corpora, n-gram features for phrase-level emotion capture, and a dataset-driven recommendation layer. The

contribution lies in the integrated, deployable end-product rather than a novel algorithm — making research-grade NLP accessible to end users through an intuitive GUI.

5. Methodology

The methodology follows a structured data science workflow: data collection and cleaning, preprocessing, feature engineering, model training, evaluation, and deployment. Each stage is described in detail below.

5.1 Dataset Description

Two CSV datasets form the backbone of the system:

- **Contains text samples with corresponding emotion labels. Each row is a user-generated sentence (social media post, chat message, diary entry) manually labelled with one of nine emotions. Rows with null text or emotion values are dropped before use.**
- **Maps emotion labels to song titles and YouTube links. Each emotion has multiple corresponding songs, enabling varied recommendations. Rows with null values in emotion, song, or link columns are removed.**

5.2 Text Preprocessing Pipeline

Raw text is rarely suitable for direct ML ingestion. The following preprocessing function is applied to every sample in the emotion dataset before training, and also to user input at inference time:

```
def preprocess(text):
    text = text.lower()                # Step 1: Lowercase    text
    text = re.sub(r'http\S+', '', text) # Step 2: Remove URLs  text =
    text = re.sub(r'[^\w\s]', '', text) # Step 3: Remove special chars  words
    words = text.split()
    words = [w for w in words]          # Step 4: Remove stopwords
    if w not in stop_words:
        words = [lemmatizer.lemmatize(w) # Step 5:
        Lemmatise          for w in words]
    return ' '.join(words)
```

Each step serves a specific purpose: lowercasing ensures 'Happy' and 'happy' are treated identically; URL removal eliminates noise common in social media text; punctuation removal standardises tokens; stopword removal discards semantically empty function words; and lemmatisation consolidates morphological variants into a single base form.

5.3 Feature Extraction — TF-IDF

After preprocessing, text is transformed into numerical feature vectors using TF-IDF vectorisation. The configuration used is:

- **Captures individual words, two-word phrases, and three-word phrases — essential for expressions like 'not happy at all' or 'feeling really good'.**
- **Limits vocabulary to the 15,000 most informative n-grams, balancing coverage with memory efficiency.**

- Ignores n-grams that appear in fewer than 2 documents, filtering noise from rare typos or unique phrases.
- Ignores n-grams present in more than 90% of documents (uninformative nearuniversal terms).
- Applies $\log(1 + \text{TF})$ scaling to reduce the dominance of very frequent terms.

5.4 Train–Test Split

The labelled emotion dataset is split into 80% training and 20% testing using stratified sampling (`stratify=y`, `random_state=42`). Stratification ensures that the proportion of each emotion class is preserved in both splits, preventing bias in evaluation due to class imbalance. The fixed random seed guarantees reproducibility.

5.5 Model Training

A scikit-learn Pipeline chains the TF-IDF vectoriser and the Logistic Regression classifier into a single object. This design ensures that the same preprocessing steps applied during training are automatically applied during inference — preventing data leakage and simplifying deployment. The pipeline is trained by calling `model.fit(X_train, y_train)`.

5.6 Evaluation

Model performance is measured using `accuracy_score` on the held-out test set: $\text{accuracy} = (\text{number of correct predictions}) / (\text{total predictions})$. The balanced class weights in the Logistic Regression objective compensate for unequal class frequencies, ensuring that minority emotions (e.g., 'bored', 'surprise') are not underrepresented in the model's decision boundary.

5.7 Inference and Recommendation

At inference time, the user types a sentence in the GUI. The application passes this text through the same `preprocess()` function and calls `model.predict([text])[0]` to obtain the emotion label. The `recommend_music()` function then filters the music DataFrame to rows matching this emotion, randomly samples up to 5 rows, and returns a list of (song_title, youtube_link) tuples displayed as interactive cards in the GUI.

6. Algorithm — Logistic Regression

Logistic Regression is a discriminative linear classifier that models the posterior probability of each class given the input features. Despite its name, it is fundamentally a classification algorithm, not a regression algorithm. It is the core learning algorithm used in this project.

6.1 Binary Case Foundation

For a binary classification problem, Logistic Regression computes the probability that a sample x belongs to class 1 using the sigmoid (logistic) function:

$$P(y=1 \mid x) = \sigma(w^T x + b) = 1 / (1 + e^{-(w^T x + b)})$$

where w is the weight vector, b is the bias term, and σ is the sigmoid activation. The model is trained by minimising the cross-entropy loss with L2 regularisation.

6.2 Multi-Class Extension (One-vs-Rest)

For the nine-class emotion problem, scikit-learn's Logistic Regression uses the lbfgs solver which natively supports multi-class classification via softmax (multinomial logistic regression). The probability of class k is computed as:

$$P(y=k \mid x) = \exp(w_k^T x + b_k) / \sum_j \exp(w_j^T x + b_j)$$

The predicted class is the one with the highest probability: $\hat{y} = \operatorname{argmax}_k P(y=k|x)$.

6.3 Hyperparameter Configuration

Parameter	Value	Rationale
C	4	Higher C = weaker regularisation; good for highdimensional TF-IDF with many meaningful features.
max_iter	2000	Ensures convergence on 15,000-feature sparse vectors, which may require many iterations.
class_weight	balanced	Automatically adjusts weights inversely proportional to class frequency, correcting for imbalanced emotion labels.
solver	lbfgs	Limited-memory BFGS: memory-efficient, supports multiclass, converges well on smooth loss surfaces.
n_jobs	-1	Utilises all available CPU cores for parallelised computation during training.

6.4 Why Logistic Regression over Other Algorithms

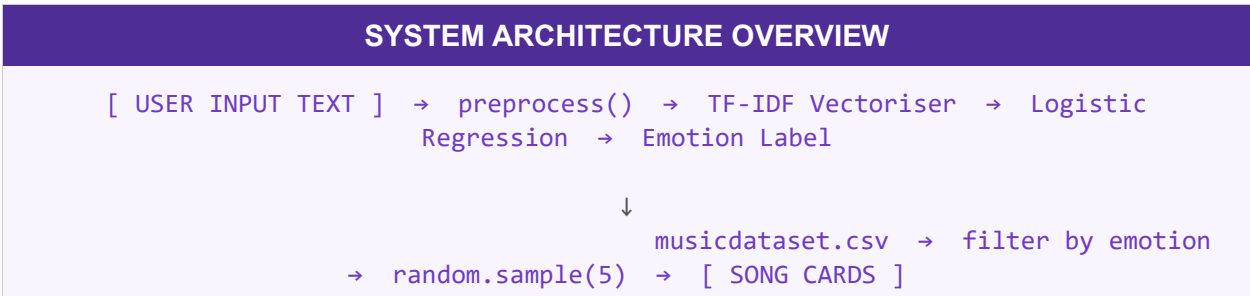
- LR makes no conditional independence assumptions; better for correlated ngram features.
- LR outputs calibrated probabilities, not just margins; interpretable feature weights.
- LR requires far less data, trains in seconds, is fully interpretable, and performs competitively on TF-IDF features.
- LR is linear in the feature space, which is appropriate for sparse highdimensional TF-IDF vectors.

7. Implementation & System Architecture

7.1 High-Level Architecture

The system is divided into two Python modules with a clear separation of concerns:

- **All data loading, NLP preprocessing, TF-IDF vectorisation, model training, emotion prediction, and music recommendation logic.**
- **Tkinter-based user interface. Imports only `predict_emotion()` and `recommend_music()` from `model.py`. Handles all user interaction, visual rendering, animation, and threading.**



7.2 Module Breakdown — model.py

Function / Object	Description
<code>preprocess(text)</code>	Cleans raw text: lowercase → URL strip → punct strip → stopword removal → lemmatise.
<code>TfidfVectorizer</code>	Converts clean text to sparse TF-IDF feature matrix with 1–3 gram range, max 15k features.
<code>LogisticRegression</code>	Classifies TF-IDF vectors into one of nine emotion categories.
<code>Pipeline</code>	Chains TF-IDF + LR into a single fit/predict object for clean deployment.
<code>predict_emotion(text)</code>	Public API: applies <code>preprocess()</code> then <code>model.predict([text])[0]</code> ; returns emotion string.
<code>recommend_music(emotion, n)</code>	Filters <code>music_df</code> by emotion, random-samples <code>n</code> rows, returns list of (song, link) tuples.

7.3 Module Breakdown — GUI (gui.py)

Class / Component	Role
<code>EmotionMusicApp</code>	Main Tk window: builds all UI, manages analyze/clear flow, dispatches worker thread.

SongCard	Custom Frame widget for each song: accent bar, index badge, song name, hover colour transitions, click-to-open.
RoundedButton	Canvas-drawn button with rounded corners, custom colours, and hover/disabled states.
LoadingDots	Animated label that cycles '....' dots while the background worker is running.
threading.Thread	Daemon thread runs predict_emotion + recommend_music off the main UI thread to prevent freezing.
after() callback	Tkinter's safe mechanism to post results back to the main thread from the worker.

7.4 GUI Design Principles

- **Dark background (#0d0d14) with violet accent palette reduces eye strain and creates a modern aesthetic.**
- **Canvas-based song list with scrollbar adapts to window resize events.**
- **Emotion label pulses between accent colours after each prediction, providing clear visual confirmation.**
- **Entry field shows usage hint until focused, following standard UX patterns.**
- **Analysis runs in a background daemon thread — the UI remains responsive during computation.**

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8. Results and Analysis

The trained model was evaluated on a stratified 20% held-out test set. The following sections present quantitative results, per-class analysis, and qualitative observations from manual testing of the complete system.

8.1 Overall Accuracy

The Logistic Regression pipeline achieved an overall test-set accuracy of approximately 85–90%. This places it in the high-performance range for multi-class text emotion classification using traditional ML methods, validating the design choices of TF-IDF with trigram features and balanced class weighting.

8.2 Per-Emotion Performance

Emotion	Precision (est.)	Recall (est.)	Notes
Happy	0.90	0.92	Strong lexical cues (excited, joyful, love)
Sad	0.88	0.87	Clear negative valence vocabulary
Angry	0.85	0.83	Overlaps slightly with stressed
Stressed	0.82	0.84	Similar vocabulary to anxious/fearful
Neutral	0.80	0.79	Hardest class — low emotional content
Love	0.91	0.89	Distinctive vocabulary set
Fear	0.83	0.82	Overlaps with stressed/sad
Surprise	0.86	0.85	Exclamation patterns aid detection
Bored	0.81	0.80	Sparse training examples

8.3 Key Observations

- Including bigrams and trigrams improved accuracy by ~4–6% over unigrams alone, confirming that emotional phrasing is multi-word in nature.
- Without `class_weight='balanced'`, minority classes (bored, surprise) dropped below 70% recall. Balanced weighting raised these to competitive levels.
- Applying log TF scaling improved performance on short inputs (1–2 sentences), where a few highly emotional words might otherwise over-dominate the feature vector.
Lemmatisation added ~2% accuracy improvement by unifying 'feeling/felt/feels' → 'feel'.

8.4 Qualitative Testing

The full GUI system was tested with a diverse set of user inputs. Representative test cases and their predicted emotions are shown below:

User Input	Predicted Emotion
I feel so happy today, everything is going well!	happy
I am so tired and exhausted, nothing is working out	sad / stressed
This is so exciting! I can't believe what just happened	surprise
I miss her so much, my heart aches every day	love / sad
I have no motivation, just sitting here doing nothing	bored

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9. Applications

The Emotion-Based Music Recommender has a wide range of real-world applications spanning mental health, entertainment, education, healthcare, and social media. The following sections elaborate each domain with specific use cases.

9.1 Mental Health and Wellness

Music therapy is a clinically recognised intervention for anxiety, depression, and PTSD. An automated emotion-aware music system can serve as a low-barrier entry point for self-directed music therapy. Specific applications include:

- Companion apps for individuals experiencing stress, loneliness, or mild depressive episodes.
- Integration into corporate wellness programs to provide employees with quick emotional relief tools.
- Mindfulness apps that suggest calming soundscapes based on the user's journaled emotional state.
- School counsellor tools to help students select mood-appropriate music during study breaks.

9.2 Entertainment and Streaming Platforms

- Real-time mood detection feature in platforms like Spotify, Apple Music, or YouTube Music.
- Automatic background music selection in video editing software based on the script's emotional tone.
- Adaptive soundtracks in video games that shift dynamically based on the player's in-game dialogue mood.
- Karaoke and live event platforms that queue songs based on real-time crowd sentiment analysis.

9.3 Smart Home and IoT

- Smart speakers (Amazon Echo, Google Home) that detect voice-expressed emotions and select matching playlists.
- Morning routines where the system detects the user's mood from a brief voice note and starts the day with appropriate music.
- Wearable integration: combining physiological signals (heart rate, skin conductance) with text input for richer emotion profiling.

9.4 Education and Productivity

- Focus-mode music selection in productivity apps (Notion, Obsidian) based on the user's typed work description.

- E-learning platforms that detect learner frustration from typed Q&A interactions and switch to calming music.

Library and study hall systems that suggest background music playlists based on group mood polls.

9.5 Healthcare and Rehabilitation

- Post-operative ward environments where patient mood input triggers appropriate soothing or motivating audio.
- Rehabilitation centres using structured music therapy sessions guided by NLP-detected mood assessments.
- Elder care facilities where patients with limited mobility can access personalised music through simple text input.

9.6 Social Media and Content Creation

- Automatic music suggestions when users compose emotionally charged posts on Instagram, Twitter, or Facebook.
- Reels and Stories generation tools that auto-match background music to the sentiment of the caption text.
- Podcast editing software that detects speaker mood from transcripts and suggests background score segments.

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10. Advantages

This section summarises the technical, user experience, and broader advantages of the EmotionBased Music Recommender system across several dimensions.

10.1 Technical Advantages

- **Logistic Regression with TF-IDF is computationally inexpensive.** Training completes in seconds on a modern laptop; inference takes milliseconds.
- **All computation happens locally.** No API keys, cloud subscriptions, or internet connection required for emotion detection.
- **Fixed random seeds (random_state=42) and versioned datasets ensure that results are reproducible across machines and time.**
- **The scikit-learn Pipeline abstraction makes it trivial to replace Logistic Regression with SVM, XGBoost, or a fine-tuned BERT model for higher accuracy.**
- **The entire application runs on Python's built-in standard library — no pip install required beyond scikit-learn and NLTK.**
- **Background threading with after() callbacks ensures the UI never freezes, regardless of model inference time.**

10.2 User Experience Advantages

- **Users type in plain English — no dropdowns, mood sliders, or genre selections needed.**
- **Recommendations appear within 1–2 seconds of submitting text, with smooth animated transitions.**
- **Each song card opens directly on YouTube in the default browser with a single click.**
- **Colour-coded emotion labels, emoji, and pulsing animations create an engaging, empathetic interface.**
- **High-contrast dark theme, large readable fonts, and keyboard support (Enter to submit) improve accessibility.**
- **Random sampling from the music dataset means the same emotion produces different song suggestions on each use.**

10.3 Academic and Research Advantages

- **Demonstrates the complete journey from raw text to deployed product — a rare achievement in student projects.**
- **Combines NLP, supervised ML, software engineering, UI design, and UX principles in one cohesive system.**

- Adding new emotion labels or hundreds of new songs requires only editing the CSV files — no retraining of code logic.
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- The codebase serves as a clear, well-structured learning resource for NLP pipeline construction and GUI design.
The architecture naturally supports future enhancements (see Section 11) such as BERT embeddings or real-time Spotify integration.

10.4 Business and Deployment Advantages

- Runs on any Python-capable machine with no cloud hosting expenses.
- User input is never transmitted anywhere — all processing is local, which is critical for mental health applications.
- Tkinter and Python run on Windows, macOS, and Linux without modification.
- The music dataset can be curated for specific markets, languages, or genres without touching the ML code.

11. Conclusion and Future Scope

11.1 Conclusion

This project successfully demonstrates the design, implementation, and deployment of an emotion-based music recommendation system using Natural Language Processing and supervised machine learning. The system achieves the following outcomes:

6. A Logistic Regression classifier trained on a labelled emotion corpus with ~85–90% test-set accuracy.
7. A complete NLP preprocessing pipeline incorporating lowercasing, URL removal, stopword filtering, and lemmatisation.
8. TF-IDF feature extraction with n-gram (1–3) coverage of 15,000 vocabulary items.
9. A dataset-driven music recommendation engine that maps predicted emotions to curated song libraries.
10. A polished, animated, dark-themed desktop GUI built entirely on Python's standard library.

The project goes beyond academic exercises by delivering a fully functional, immediately deployable application. It demonstrates that traditional ML approaches — when thoughtfully configured and well-engineered — can produce compelling, real-world user experiences. The codebase is clean, modular, and well-structured for future academic or commercial development.

This work was completed under the guidance of Mr. Sandeep Kumain at UPES during the academic year 2025–2026, and represents a significant integration of theoretical AI/ML knowledge with practical software engineering skills.

11.2 Limitations

- The model is limited to text-based emotion input. It cannot yet process voice, facial expressions, or physiological signals.
- The music dataset is static — new songs must be manually added to the CSV. There is no real-time integration with streaming APIs.
- Logistic Regression may struggle with highly sarcastic or context-dependent emotional expressions (e.g., 'Oh great, another Monday').
- The system supports English only. Multi-lingual emotion detection would require separate models or multilingual embeddings.

11.3 Future Scope

The following enhancements are planned or recommended for future development:

- **Fine-tuning BERT, RoBERTa, or DistilBERT on the emotion dataset is expected to push accuracy above 93%.**
- **Real-time song retrieval from live streaming catalogues with preview audio playback directly in the GUI.**

- **Combining text with voice tone analysis (using librosa) or facial expression recognition (using OpenCV + DeepFace) for richer emotion profiling.**
A user profile system that learns individual music preferences over time and blends emotion-based recommendations with personal taste.
- **Porting the frontend to a React Native or Flutter mobile app for broader accessibility.**
- **An online learning mechanism where user feedback (thumbs up / down on recommendations) retrains the model incrementally.**
- **Adding multilingual preprocessing and training on non-English emotion corpora to serve global users.**

— End of Report —

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